

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Technology (CSE)

QUIZ #1 (SET-SAMPLE)
DURATION: 25 MINUTES

SUMMER SEMESTER, 2024-2025
FULL MARKS: 20

CSE 4621: Machine Learning

Programmable calculators are not allowed.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID:

1. What is the definition of a convex set?

5
(CO1)
(PO1)

Solution:

A set C is convex if for any two points $x, y \in C$ and any $t \in [0, 1]$, the point $tx + (1 - t)y$ also lies in C . In plain terms, the straight line segment between any two points in the set must lie entirely within the set.

Rubric:

- 5 points for correct definition.

2. In Bayes' Theorem, what do the terms prior, likelihood, and posterior refer to?

5
(CO1)
(PO1)

Solution:

Prior $P(H)$ — our belief about the hypothesis before seeing data. Likelihood $P(D|H)$ — probability of observing the data given the hypothesis. Posterior $P(H|D)$ — updated belief about the hypothesis after observing the data.

Rubric:

- 1.67 points for correctly stating each segment.

3. Explain why convexity is a desirable property for a machine learning cost function.

5
(CO1)
(PO1)

Solution:

A convex cost function guarantees that any local minimum found is also the global minimum. Gradient descent will reliably converge to the best possible solution regardless of the starting point. When the cost function is non-convex, gradient descent can get stuck at local minima or saddle points, meaning the model may converge to a suboptimal solution depending on initialization.

Rubric:

- 5 points for correctly explaining.

4. For a univariate linear regression model $h_{\theta}(x) = \theta_0 + \theta_1 x$ with training data $\{(1, 1), (2, 2), (3, 3)\}$, compute $J(\theta)$ when $\theta_0 = 0$ and $\theta_1 = 0.5$. 5
(CO1)
(PO1)

Solution:

$$J(\theta) = \frac{1}{2 \times 3} (0.25 + 1 + 2.25) = 0.583$$

Rubric:

- 5 points for correctly finding the loss.